like FinFET and GAA, and the key challenge is ensuring that all foundry-specific design rules are correctly handled by the tools during implementation.

At the same time, major demand for EDA solutions comes from highly complex, cutting-edge designs, both in India and globally. These include Arm-based systems, which are becoming ever more complex, and high-speed interfaces such as universal chiplet interconnect express (UCIe) and advanced memory interfaces that need very fast data transfer. Such systems also involve building subsystems with multiple IPs and integrating them with processors—sometimes 20, 30, or even more—while running different workloads. The challenge, therefore, is not just about advanced technology nodes but also about managing increasing system-level complexity, integration, and verification across the full design stack.

On the chiplet era

EFY: Will chiplets eventually replace large monolithic systems on chips (SoCs) for most high-performance applications?

SS: Yes, in many high-performance applications, chiplets are expected to overtake large monolithic SoCs. There is also work on wafer-scale approaches, where an entire wafer is treated as a single system, especially in data centre use cases. The shift is mainly driven by reuse and the ability to scale product lines more efficiently. However, the choice still depends on whether a customer wants a multi-die solution or a full wafer-scale chip, since wafer-scale systems are expensive to handle and bring several second-order system effects. Overall, for domains like data centres, the trend is clearly towards multi-die chiplet-based designs rather than large monolithic SoCs.

EFY: How are advanced packaging technologies changing the relationship between design and manufacturing?

SS: Advanced packaging technologies have made design and manufacturing much more interconnected. Today, it is a continuous 'correct-by-construction' process in which both teams exchange data and models throughout development. Design teams provide specific formats and information to packaging engineers, while packaging teams supply models that help designers perform analysis and optimisation. Some improvements are made at the design stage, while others are handled during packaging.

> **"DESIGN AND MANUFACTURING ARE NO LONGER ISOLATED FUNCTIONS."**

Because packaging decisions affect factors such as form factor, pin count, current-carrying capacity, and overall system requirements, packaging must be considered very early in the development cycle, often when the specification or RTL is being defined. The package architecture needs to be largely finalised before many design decisions can be completed. As a result, design and manufacturing are no longer isolated functions; they operate as an integrated process, with close collaboration from the beginning of the project.

EFY: What verification challenges emerge when moving from single-die design to chiplet-based architectures?

SS: The main challenge is ensuring correct system-level behaviour across the full chiplet architecture, including power-up sequencing, high-speed interfaces, and workload execution for cloud-level use cases. To address this, hardware-assisted verification (HAV) is used, in which the full chip is modelled on platforms like ZeBu before silicon is available. This allows customers to run real workloads pre-silicon, including complex AI and large language models (LLMs) as well as smaller software workloads, and observe system behaviour early. The process helps identify issues and guide design improvements before tape-out, and continues iteratively until the chip is ready. Once silicon is available, the same workloads can be run again on the actual hardware to confirm real-world performance and correctness.

On the engineer's toolkit

EFY: What new skills should semiconductor engineers develop to work effectively with AI-enabled design tools?

SS: From an engineer's point of view, strong fundamentals remain essential. There is no shortcut around understanding how transistors work, how digital design fundamentals and logic gates operate, and what design trade-offs need to be addressed. At the same time, engineers should avoid becoming purely task-focused and instead develop knowledge of protocols and systems. Since chips ultimately go into applications such as mobile devices, surveillance cameras, and other intelligent systems, engineers need to understand the broader system context and how a design can be improved for its end use. Innovation increasingly happens across domains, which is why fundamentals, systems thinking, and protocol knowledge matter.

The approach to workforce development is also changing. Rather than simply learning how to use a tool, engineers should focus on what they want to achieve with it. The key question is not how to use the tool, but what problem they are trying to solve with it—a mindset that all engineers should adopt. EFY